

Challenge (Habanero)

- Q1.** A recipe for cake requires $3 \cdot x + 2$ cups of flour and $2 \cdot x - 1$ cups of sugar. If $x = 2$, how many cups of flour and sugar are needed in total?
 (A) 12
 (B) 10
 (C) 14
 (D) 16
 (E) 18
- Q2.** A builder needs to mix $2x + 5$ bags of concrete and $x - 3$ bags of sand for a construction project. If $x = 4$, how many bags of concrete and sand are needed in total?
 (A) 15
 (B) 12
 (C) 18
 (D) 10
 (E) 20
- Q3.** A chef is making a sauce that requires $\frac{2}{3} \cdot x + 1$ cups of water and $\frac{1}{2} \cdot x - 1$ cups of oil. If $x = 6$, how many cups of water and oil are needed in total?
 (A) 6
 (B) 5
 (C) 7
 (D) 4
 (E) 8
- Q4.** An architect is designing a building with $5 \cdot x - 2$ floors and each floor has $3 \cdot x + 1$ rooms. If $x = 3$, how many rooms are there in total?
 (A) 120
 (B) 130
 (C) 140
 (D) 150
 (E) 160
- Q5.** A recipe for cookies requires $\frac{3}{4} \cdot x + 2$ cups of butter and $\frac{2}{3} \cdot x - 1$ cups of sugar. If $x = 8$, how many cups of butter and sugar are needed in total?
 (A) 11
 (B) 12
 (C) 13
 (D) 14
 (E) 15
- Q6.** A contractor needs to lay $4x + 1$ meters of pipe and $2x - 3$ meters of wire for a construction project. If $x = 5$, how many meters of pipe and wire are needed in total?
 (A) 30
 (B) 32
 (C) 34
 (D) 36
 (E) 38
- Q7.** A chef is making a dish that requires $2 \cdot x + 3$ cups of flour and $x - 2$ cups of sugar. If $x = 6$, how many cups of flour and sugar are needed in total?
 (A) 16
 (B) 14
 (C) 18
 (D) 12
 (E) 20
- Q8.** An architect is designing a bridge with $3 \cdot x - 1$ spans and each span has $2 \cdot x + 2$ supports. If $x = 4$, how many supports are there in total?
 (A) 80
 (B) 70
 (C) 90
 (D) 60
 (E) 100

Open Ended Questions (FRQ)

- Q9.** A contractor is building a house that requires $x + 2$ loads of lumber and $3x - 1$ loads of roofing material. If $x = 5$, calculate the total number of loads of lumber and roofing material needed and show your solution.

- Q10.** A chef is making a recipe that requires $2x + 1$ cups of flour and $x - 3$ cups of sugar. If $x = 8$, calculate the total number of cups of flour and sugar needed and simplify the expression $2x + 1 + x - 3$.

Chef's Tips

- Check calculations carefully
- Ensure all questions are answered
- Use a calculator only when necessary